

WELL COMPLETION REPORT (SYNTHETIC)

OIL-DGB-004 - Digboi-004

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000037
Field / District	Digboi / Tinsukia
Block	PEL-ASSAM-01
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	2021-02-18 / 2021-06-30
Measured / TVD	2558 m / 2525 m
Trajectory	Vertical; inc 1.0 deg; az 259.5 deg
Formation Interval	1304-2009 m
Lithology	Sandstone-siltstone sequence
Pore Pressure Reference	2485 psi
Fracture Gradient	15.25 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	Low (35.0/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	639	Surface
9 5/8 in casing	1483	Intermediate
7 in liner/casing	2251	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2021-04-12 1155 m	NPT	High	ROP dropped to zero during equipment unavailability. Mitigation: Classify NPT cause consistently for offset-well learning. Lesson: Consistent NPT coding enables meaningful cross-well comparison.
2021-05-25 1922 m	Torque Spike	High	High torque developed while WOB remained elevated. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2021-05-27 1970 m	Kick	High	Pit gain and flow with pumps off were observed. Mitigation: Increase mud weight within fracture-margin limits and monitor gas trend. Lesson: Maintain mud-weight window using offset pressure evidence.
2021-06-08 2180 m	Mud Loss	High	Partial returns observed at shale shaker. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Track flow balance and pit volume trend before losses become severe.
2021-04-30 1474 m	Stuck Pipe	Medium	High drag was recorded while circulating remained possible. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency.
2021-05-11 1670 m	Torque Spike	Medium	Sharp torque excursion occurred with RPM instability. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2021-05-28 1979 m	Mud Loss	Medium	Partial returns observed at shale shaker. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Track flow balance and pit volume trend before losses become severe.

Completion Summary

The synthetic well reached 2558 m MD in the Tipam interval. The NWIS knowledge record links 12 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.